Program 2: Maven Project with Guava and Commons-IO

Dependencies Create a Maven Project Using Terminal by following command:

mvn archetype: generate -DgroupId=com.example -DartifactId=MyMavenGuavaApp -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false

```
sumanth@sumanth-VirtualBox:~/devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenGuavaApp -DarchetypeArtifactI
d=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/sumanth/devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenGuavaApp
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/sumanth/devops/MyMavenGuavaApp
[INFO] ----------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ----------------------------------------------------------------------------
[INFO] Total time:  1.404 s
[INFO] Finished at: 2026-03-19T13:32:06+05:30
[INFO] ----------------------------------------------------------------------------
sumanth@sumanth-VirtualBox:~/devops$
```

Now Navigate to the Project Directory using **cd MyMavenGuavaApp**

Edit the pom.xml File and add the code

```
Open   *pom.xml   ~/Devops/MyMavenGuavaApp   Save
1  <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
2    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
3    <modelVersion>4.0.0</modelVersion>
4    <groupId>com.example</groupId>
5    <artifactId>MyMavenGuavaApp</artifactId>
6    <packaging>jar</packaging>
7    <version>1.0-SNAPSHOT</version>
8    <name>MyMavenGuavaApp</name>
9    <url>http://maven.apache.org</url>
10   <dependencies>
11     <dependency>
12       <groupId>junit</groupId>
13       <artifactId>junit</artifactId>
14       <version>3.8.1</version>
15       <scope>test</scope>
16     </dependency>
17     <!-- https://mvnrepository.com/artifact/com.google.guava/guava -->
18 <dependency>
19     <groupId>com.google.guava</groupId>
20     <artifactId>guava</artifactId>
21     <version>33.4.0-jre</version>
22 </dependency>
23 <!-- https://mvnrepository.com/artifact/commons-io/commons-io -->
24 <dependency>
25     <groupId>commons-io</groupId>
26     <artifactId>commons-io</artifactId>
27     <version>2.18.0</version>
28 </dependency>
29   </dependencies>
30   <!-- Build: Configuring plugins and build settings -->
31 <build>
32 <plugins>
33 <!-- Example: Maven Compiler Plugin to compile Java code -->
34 <plugin>
35 <groupId>org.apache.maven.plugins</groupId>
36 <artifactId>maven-compiler-plugin</artifactId>
37 <version>3.8.1</version>
38 <configuration>
```

```
</plugin>
<!-- Example: Maven Surefire Plugin to run tests -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-surefire-plugin</artifactId>
<version>2.22.2</version>
</plugin>


<plugin>
            <groupId>org.apache.maven.plugins</groupId>
            <artifactId>maven-jar-plugin</artifactId>
            <version>3.1.0</version>
            <configuration>
                <archive>
                    <manifestEntries>
                        <Main-Class>com.example.App</Main-Class>
                    </manifestEntries>
                </archive>
            </configuration>
        </plugin>

</plugins>
</build>
</project>
```

Modify App.java

Got to App.java using **cd src/main/java/com/example/** and **gedit App.java** to type the file with following code:

```
Open                App.java                                                Save
                    ~/Devops/MyMavenGuavaApp/src/main/java/com/example
package com.example;
import org.apache.commons.io.FileUtils;
import java.io.File;
import java.io.IOException;
import com.google.common.collect.ImmutableList;

public class App
{

    public static void main( String[] args )
    {
        // Create an immutable list of strings (fruits)
        // This list cannot be changed (no add/remove allowed)
        ImmutableList<String> fruits = ImmutableList.of("Apple", "Banana", "Cherry");

        // Print the list of fruits to the console
        System.out.println(fruits);

        // Create a File object representing the source file
        File sourceFile = new File("source.txt");

        // Create a File object representing the destination file
        File destFile = new File("destination.txt");

        try {
            // Copy the contents of source.txt to destination.txt
            // Using Apache Commons IO FileUtils class
            FileUtils.copyFile(sourceFile, destFile);

            // Print success message if file is copied
            System.out.println("File copied successfully!");
        }
        catch (IOException e) {

            System.err.println("Error occurred while copying file: " + e.getMessage());
        }
    }
}----
```

Build the Project using the command **mvn clean install** command

```
(base) sumanth@sumanth-VirtualBox:~/devopss6d/MyMavenGuavaApp$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.example:MyMavenGuavaApp >--------------------
[INFO] Building MyMavenApp 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MyMavenGuavaApp ---
[INFO] Deleting /home/paramanand/devops6d/MyMavenGuavaApp/target
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenGuavaApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform depe
[INFO] skip non existing resourceDirectory /home/paramanand/devops6d/MyMavenGuavaApp/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyMavenGuavaApp ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/paramanand/devops6d/MyMavenGuavaApp/target/classes
[INFO]
[INFO] --- maven-resources-plugin:3.6:testResources (default-testResources) @ MyMavenGuavaApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform depe
[INFO] skip non existing resourceDirectory /home/paramanand/devops6d/MyMavenGuavaApp/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:testCompile (default-testCompile) @ MyMavenGuavaApp ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/paramanand/devops6d/MyMavenGuavaApp/target/test-classes
[INFO]
[INFO] --- maven-surefire-plugin:2.22.2:test (default-test) @ MyMavenGuavaApp ---
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.008 s - in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- maven-jar-plugin:3.1.0:jar (default-jar) @ MyMavenGuavaApp ---
[INFO] Building jar: /home/paramanand/devops6d/MyMavenGuavaApp/target/MyMavenGuavaApp-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MyMavenGuavaApp ---
[INFO] Installing /home/paramanand/devops6d/MyMavenGuavaApp/target/MyMavenGuavaApp-1.0-SNAPSHOT.jar to /ho
[INFO] Installing /home/paramanand/devops6d/MyMavenGuavaApp/pom.xml to /home/paramanand/.m2/repository/com
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.418 s
[INFO] Finished at: 2026-05-17T18:03:27Z
[INFO] ------------------------------------------------------------------------
```

This compiles the project and creates a JAR in the target/ folder.

Run the Application:

If java-jar command give an error which is most probable, use the following command **mvn exec:java -Dexec.mainClass=”com.example.App”**

```
(base) sumanth@sumanth-VirtualBox:~/devops6d/MyMavenGuavaApp$ mvn exec:java -Dexec.mainClass="com.example.App"
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------- < com.example:MyMavenGuavaApp > -------------------
[INFO] Building MyMavenApp 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ] --------------------------------
[INFO]
[INFO] --- exec-maven-plugin:3.6.3:java (default-cli) @ MyMavenGuavaApp ---
[Apple, Banana, Cherry]
File copied successfully!
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  0.657 s
[INFO] Finished at: 2026-03-17T18:03:53Z
[INFO] ------------------------------------------------------------------------
(base) sumanth@sumanth-VirtualBox:~/devops6d/MyMavenGuavaApp$
```

This command gives error as

Error while copying file:Source ‘source.txt’ does not exist

Now, Create a Sample File to copy the content to destination file

```
Open    source.txt
        ~/Devops/MyMavenGuavaApp    Save
1 hoiiiiiiiiii
```

Now Run the command **mvn exec:java -Dexec.mainClass=”com.example.App**” again

```
(base) sumanth@sumanth-VirtualBox:~/devops6d/MyMavenGuavaApp$ mvn exec:java -Dexec.mainClass="com.example.App"
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< com.example:MyMavenGuavaApp >-------------------
[INFO] Building MyMavenApp 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- exec-maven-plugin:3.6.3:java (default-cli) @ MyMavenGuavaApp ---
[Apple, Banana, Cherry]
File copied successfully!
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  0.657 s
[INFO] Finished at: 2026-03-17T18:03:53Z
[INFO] ------------------------------------------------------------------------
(base) sumanth@sumanth-VirtualBox:~/devops6d/MyMavenGuavaApp$
```

The output shows file copied successfully.

We can see newly created **destination.txt** using gedit destination.txt if you are still in Project’s directory

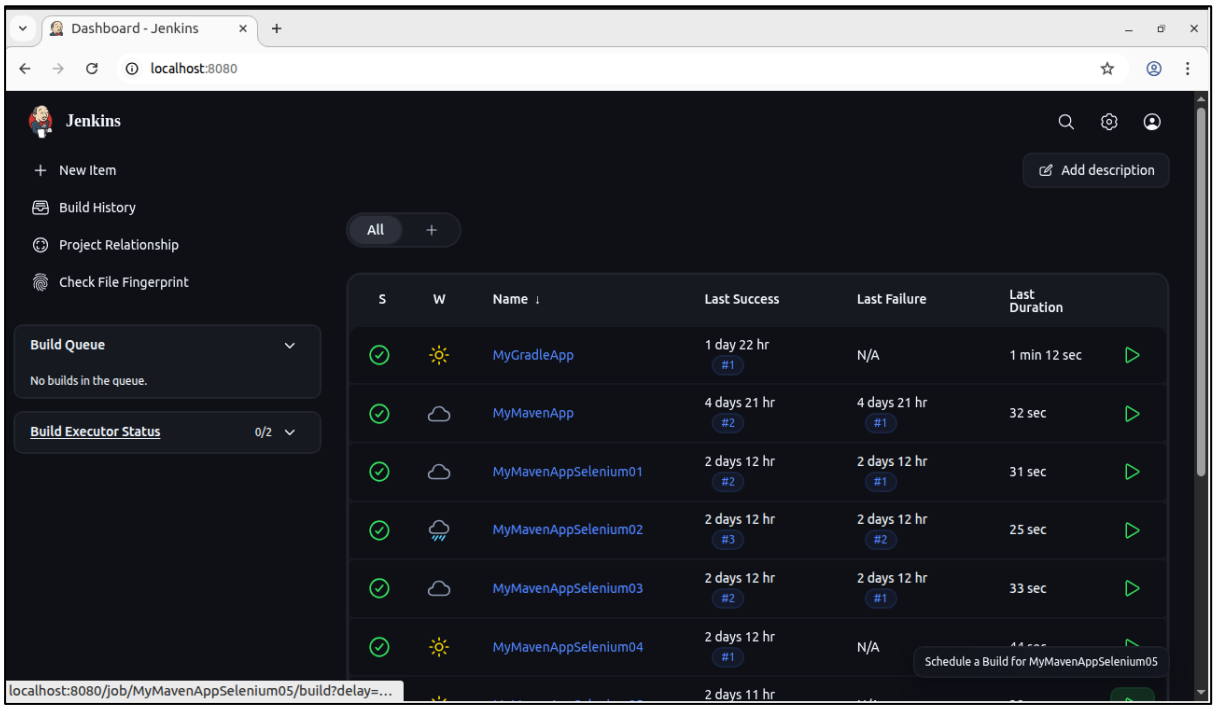
```
sumanth@sumanth-VirtualBox:~/devops/MyMavenGuavaApp$ ls
pom.xml  source.txt  src  target
sumanth@sumanth-VirtualBox:~/devops/MyMavenGuavaApp$
```

File content is same as the content in source file.

```
Open                destination.txt                Save
                    ~/Devops/MyMavenGuavaApp
1 hoiiiiiiiiii
```

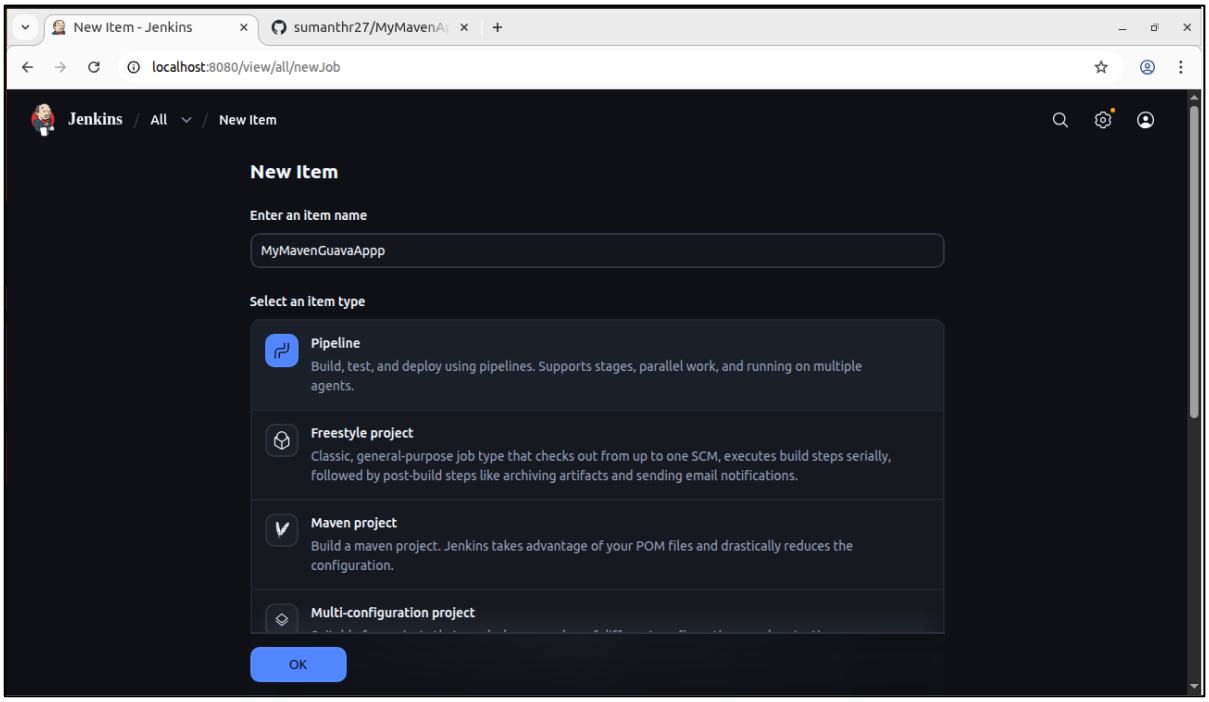
Jenkins Pipeline for MyMavenGuavaApp

Step 1: Open the Jenkins Dashboard in order to create a new item

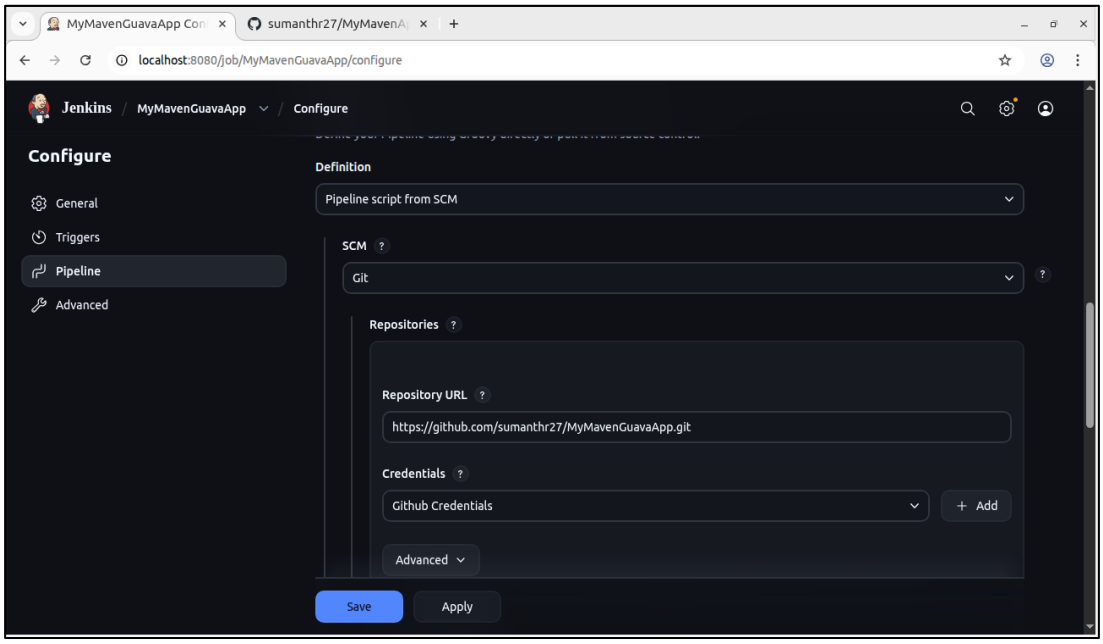


Step 2: Click on New Item in order for creating an New Item and thus give the name as “MyMavenGuavaApp” and Item type as Pipeline and

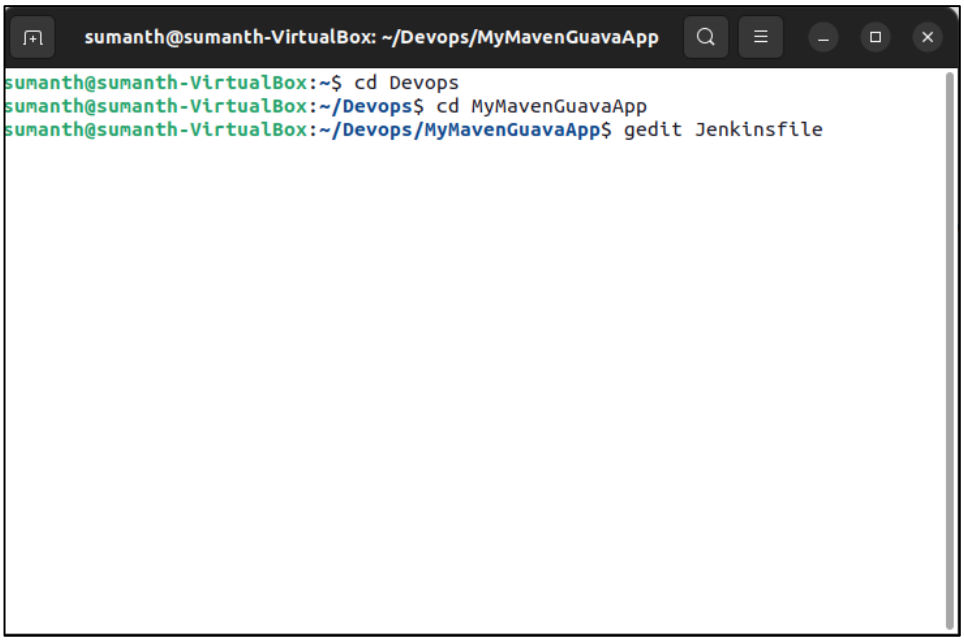
Create it



Step 3: Configure the settings by Defining the script as “Pipeline Script from SCM” and SCM as “Git”. Thus, now as we have selected Git, we need to paste the Repository URL of the project given in GitHub and choose Credentials as “GitHub Credentials” which was created earlier.



Step 4: Now in Terminal, go to directory created for MyMavenGuavaApp and create an new file called “Jenkinsfile”



Step 5: Paste the available code inside the Jenkinsfile

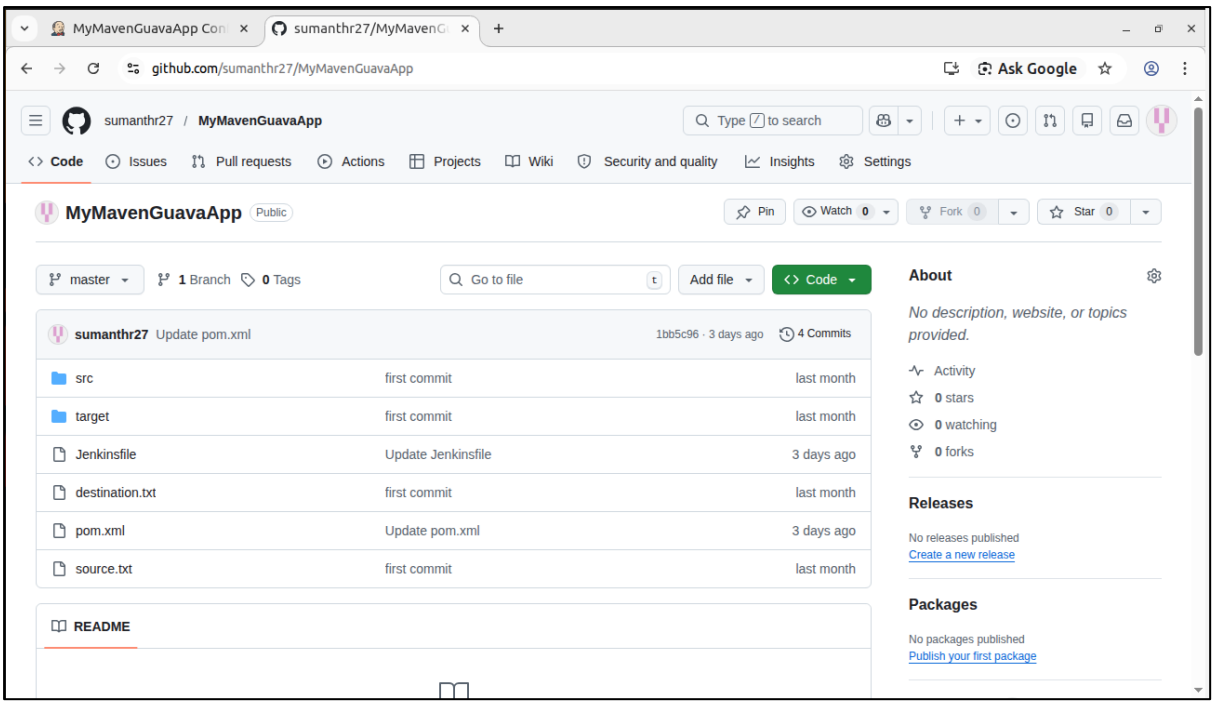
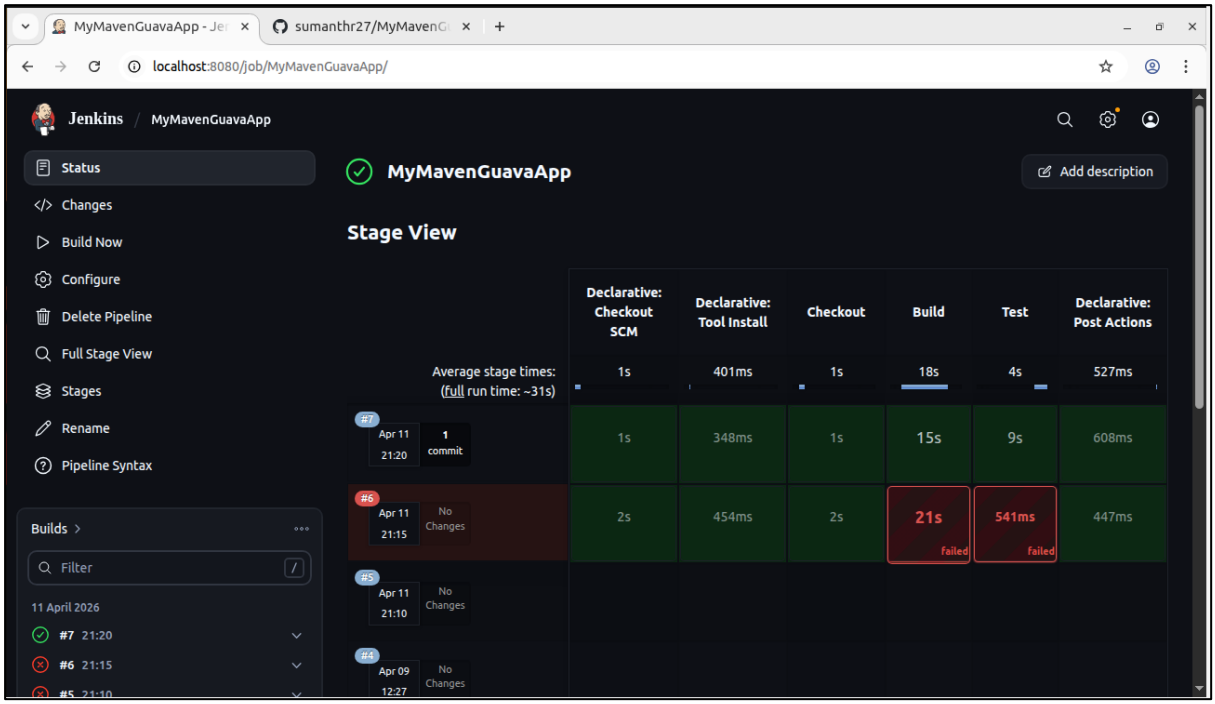
```
pipeline {
    agent any  // Use any available agent
    tools {
        maven 'Maven'  // Ensure this matches the name configured in Jenkins
    }
    stages {
        stage('Checkout') {
            steps {
                git branch: 'master', url: 'https://github.com/sumanthr27/MyMavenGuavaApp.git'
            }
        }ss
        stage('Build') {
            steps {
                sh 'mvn clean package'  // Run Maven build
            }
        }
        stage('Test') {
            steps {
                sh 'mvn test'  // Run unit tests
            }
        }
        stage('Run Application') {
            steps {
                // Start the JAR application
                sh 'java -jar target/MyMavenApp-1.0-SNAPSHOT.jar'
            }
        }
    }

    post {
        success {
            echo 'Build and deployment successful!'
        }
        failure {
            echo 'Build failed!'
        }
    }
```

Step 6: We need to update pom.xml also in order to change the versions of source and target from 1.7 to 1.8

```
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.8</source>
<target>1.8</target>
</configuration>
</plugin>
```

Step 7: Now we need to Push changes and commit by pushing the necessary "Jenkinsfile" to the respective Repo



Step 8: Now go to Jenkins and Build the Pipeline and thus check the Stage view for the successful execution.